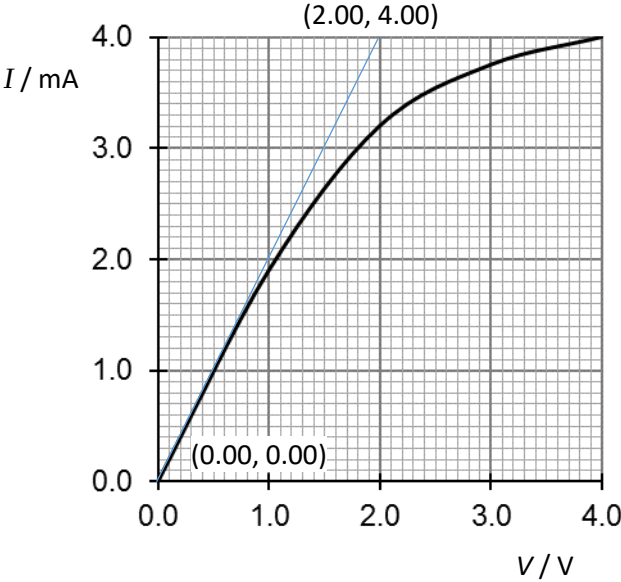


6(a)	As V increases, I increases initially. Further increase of V will cause a less than proportionate increase in I Since resistance is the ratio of potential difference to current, the resistance of the filament lamp increases.	B1 B1
	<i>Marker's comments: Common mistakes stated by candidates that the resistance is the reciprocal of the gradient of the I-V graph.</i>	

6(b)	 The minimum resistance can be determined by drawing the tangent of the graph at the origin. $\text{gradient of tangent} = \frac{(4.00 - 0.00) \times 10^{-3}}{2.00 - 0.00} = 2.00 \times 10^{-3}$ $\text{Hence, resistance} = \frac{1}{2.00 \times 10^{-3}} = 500 \, \Omega$	M1 C1 A1
6(c)	From Fig. 6.1, when $V = 4.00 \, \text{V}$, $I = 4.00 \, \text{mA}$ $\text{resistance of wire } R = \frac{4.00}{4.00 \times 10^{-3}} = 1000 \, \Omega$ $\text{resistivity } \rho = \frac{RA}{L}$ $= \frac{(1000) \left(\frac{\pi}{4} \times (0.046 \times 10^{-3})^2 \right)}{2.0}$ $= 8.3 \times 10^{-7} \, \Omega \, \text{m}$	M1 C1 A1
6(d)	From Fig. 6.1, when $I = 1.0 \, \text{mA}$, $V = 0.500 \, \text{V} = \text{terminal p.d. across cell}$ $E - Ir = 0.500$ $E - (1/1000)(0.50) = 0.500$ $E = 0.5005 \, \text{V}$	C1 A1

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7(a)	electron	B1
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